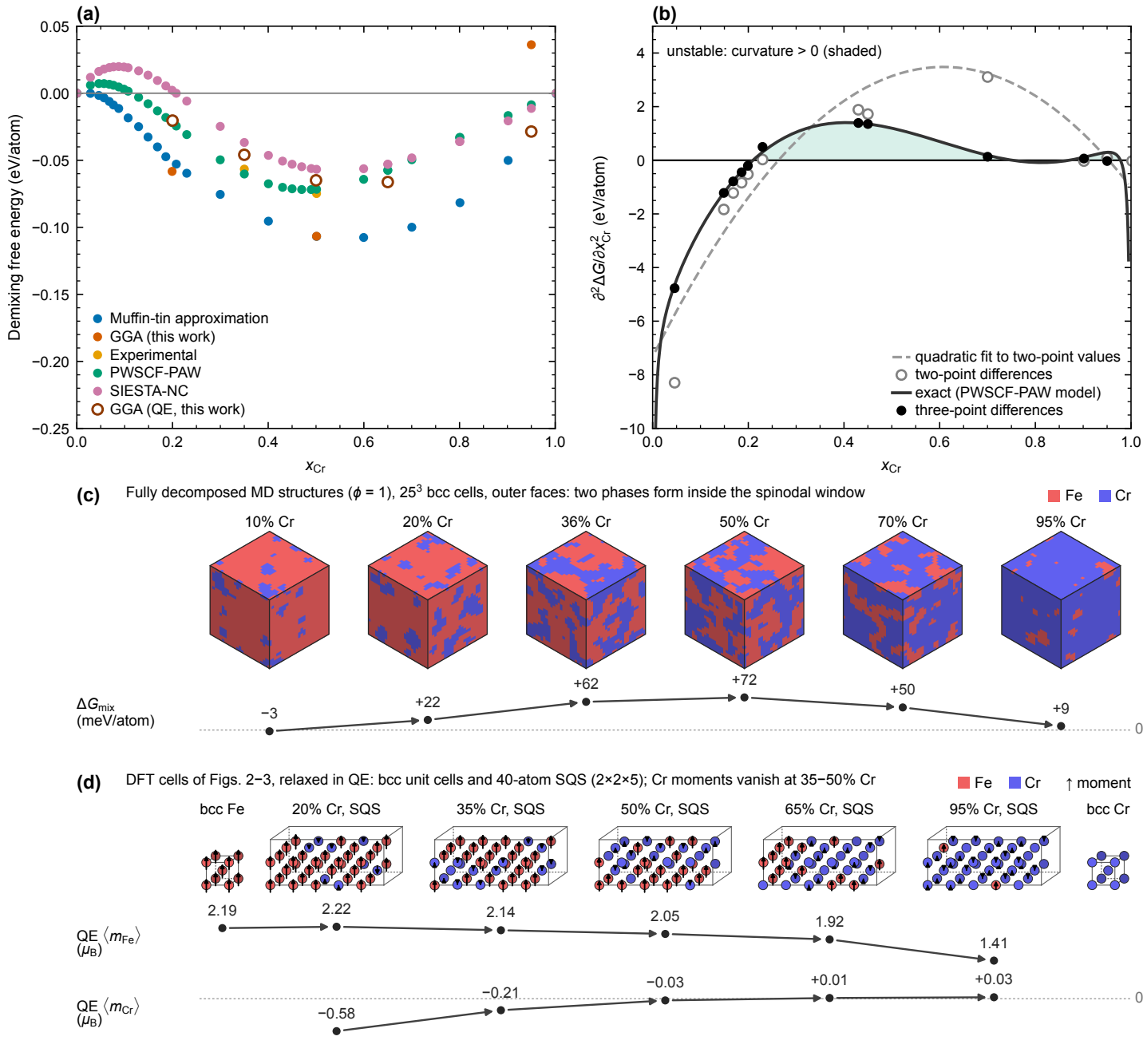

(a)
Demixing free energy (eV/atom)
0.05
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
0.0
0.2
0.4
0.6
0.8
1.0
x_{Cr}
Muffin-tin approximation
GGA (this work)
Experimental
PWSCF-PAW
SIESTA-NC
GGA (QE, this work)
(b)
unstable: curvature > 0 (shaded)
$\partial^2 \Delta G / \partial x_{Cr}^2$ (eV/atom)
4
2
0
−2
−4
−6
−8
−10
quadratic fit to two-point values
two-point differences
exact (PWSCF-PAW model)
three-point differences
(c)
Fully decomposed MD structures (ϕ = 1), 25^3 bcc cells, outer faces: two phases form inside the spinodal window
Fe
Cr
10% Cr
20% Cr
36% Cr
50% Cr
70% Cr
95% Cr
ΔG_{mix} (meV/atom)
−3
+22
+62
+72
+50
+9
0
(d)
DFT cells of Figs. 2−3, relaxed in QE: bcc unit cells and 40-atom SQS (2×2×5); Cr moments vanish at 35−50% Cr
Fe
Cr
↑ moment
bcc Fe
20% Cr, SQS
35% Cr, SQS
50% Cr, SQS
65% Cr, SQS
95% Cr, SQS
bcc Cr
QE $\langle m_{Fe} \rangle$ (μ_B)
2.19
2.22
2.14
2.05
1.92
1.41
QE $\langle m_{Cr} \rangle$ (μ_B)
−0.58
−0.21
−0.03
+0.01
+0.03
0